



## A new understanding of raindrop shape

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### ABSTRACT

In this paper raindrop shapes from laboratory, field and model investigations are examined in order to distill a consistent picture of raindrop axis ratios as a function of size. The first realistic measurements of raindrop shape were made in the UCLA vertical wind tunnel that was built under the supervision of Hans R. Pruppacher at UCLA in 1968.

Subsequent research has investigated raindrop shape in more detail by laboratory studies, field measurements and modeling. Measurements of rain at night in backscattered light have revealed that raindrops larger than 1 mm diameter oscillate in conditions where winds and collisions are too weak to excite oscillations. One apparent source of these oscillations is a resonance with eddy shedding. Subsequent laboratory studies have demonstrated that such oscillations increase the average axis ratio of small raindrops.

Details of past theoretical research show that raindrop axis ratios decrease with size in the manner of a sessile drop because of its increasing weight, but also because of an increasing differential in the fore-aft aerodynamic pressure. Modern analytical and numerical models predicted similar axis ratios for water drops falling at terminal velocity and these results are generally consistent with laboratory and field measurements.

Although the axis ratios first measured in the UCLA wind tunnel and the more recent laboratory measurements have led to a broader understanding of raindrop axis ratios, new information about raindrop shape is available from recent field measurements using improved techniques. Raindrop axis ratios, orientation and canting angles using 2-D video disdrometer (2DVD) are also reported. The results include measurements from an artificial rain experiment conducted under calm wind conditions where the drops (over 115,000) were allowed to fall a distance of 80 m. Axis ratios and drop shapes show good agreement with the Mainz wind-tunnel data and images, as well as canting angle distributions symmetric about zero degrees with a standard deviations of 7–8° for all drops > 2 mm.

Measurements of 2DVD raindrop shape has been collected from Okinawa, Sumatra, Toronto, Huntsville and Brisbane. An unusually wide axis ratio distribution was observed in the  $D = 1.6$ – $1.8$  mm region, which may be evidence for drop oscillations caused by eddy-shedding. Some subtle differences can also be seen for larger drops (e.g.  $D = 4$  mm), the vertical chord being slightly larger than the horizontal chord leading to slightly larger axis ratios compared with the 80 m fall data.

Additionally, an example using individual drop information from 2DVD measurements is considered, which enhances agreement with C-band polarimetric radar measurements. In this particular case, a considerable shift in the axis ratio distribution for 3.5 mm drops is observed during the initial strong convective phase of the event. Further ongoing work will combine the 2DVD measurements with polarimetric radar observations in a synergistic manner in order to identify the meteorological or microphysical conditions under which significant changes in drop shapes may occur in natural rain.

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Improved polarimetric radar estimates of canting angles are reported from an S-band system with high cross-polar performance antenna. The histogram of the standard deviation derived from the radar data obtained in light stratiform rain event with embedded convection shows a canting angle mode of about  $7^\circ$ , with a significant positive skewness. An increase in stability with drop diameter is also inferred from the radar measurements, which is in agreement with direct estimates from the 2DVD. Ongoing experiments in the Mainz wind tunnel are evaluating how the raindrop shapes are affected by collisions with drizzle drops within rain shafts.

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## 1. Introduction

The modern understanding of raindrop shape began with the measurements of Pruppacher and Beard (1970) using water drops suspended in the vertical air stream of the low-turbulence, wind tunnel in the UCLA Cloud Physics Laboratory. Five years later the raindrop shapes from this study were used by Seliga and Bringi (1976) in their seminal paper “Potential use of differential reflectivity at orthogonal polarization for measuring precipitation.” Research in the intervening decades resulted in considerable evidence that raindrops often have other shapes resulting from natural oscillations. This paper will review and summarize the current understanding of raindrop shape and its application in radar sensing of rainfall characteristics.

The improved axis ratio measurements in the UCLA wind tunnel were the result of its low turbulence level and a steady uniform vertical velocity across the test section that was achieved by using suction and a sonic control valve on the downstream side of the test section. A drop that was injected into the test section was maintained at a fixed vertical location by gradually reducing the tunnel velocity as the drop slowly evaporated in nearly saturated air. For the measurements of drop shape, a suspended drop was photographed against a ground-glass background illuminated by a strobe light. The photographs from Pruppacher and Beard (1970) for the larger sizes are shown in Fig. 1. The size of a drop was determined from its projected cross section by graphical integration.

The equilibrium shape of raindrops is most simply described by the axis ratio, a ratio of the maximum vertical and horizontal chords. In the measurements of Pruppacher and Beard, drops of the equivalent spherical radius  $a_0 \sim 0.5$ – $4.5$  mm (1 to 9 mm diameter) were found to have a linearly decreasing axis ratio ( $\alpha$ ) as a function

$$\alpha = 1.030 - 0.124a_0. \quad (1)$$

This equation differed substantially from Spilhaus (1948), based on spheroid shapes and their drag coefficients. It also differed from the wind tunnel measurements of Blanchard (1950) where the test section was open and the air flow was produced by a blower.

A plot of the axis ratios vs. drop radius from the results of Pruppacher and Beard and earlier data is shown in Fig. 2. The linear portion of line 5 corresponds to Eq. (1). Note that previous data apply to drops of radii  $a_0 > 1$  mm, i.e., raindrops with  $D > 2$  mm. The axis ratio departs from Eq. (1) for drizzle drops for  $a_0 < 0.25$  mm ( $D < 0.5$  mm).

In the initial experiment in the UCLA wind tunnel the fall speed of small water drops from  $a_0 = 20$   $\mu$ m to  $0.45$  mm was measured to evaluate the drag coefficients of drops for comparison to spheres (Beard and Pruppacher, 1969). The new data for larger drops resulted in drag coefficients and calculated terminal velocities in agreement with the measurements of Gunn and Kinzer (1949).

Subsequently Pruppacher and Pitter (1971a) published a semi-empirical paper on the shape of falling raindrops by adapting a perturbation method developed by Savic (1953). Their axis ratios calculations were in good agreement with the measurements of Pruppacher and Beard.

## 2. Observations

Other wind tunnel measurements around the time of Pruppacher's studies found that drop shapes varied because of oscillations (Brook and Latham, 1968, Musgrove and Brook, 1975). The oscillations occurred near the fundamental frequency with some evidence of a few higher harmonics at greatly reduced amplitude. Small drops oscillated in the most symmetric manner whereas larger drops, whose equilibrium shapes are notably different from a sphere, exhibited characteristics of the three possible modes at the fundamental frequency. Because of the orienting effect of gravity, the oscillation modes for the fundamental harmonic may be described as the axisymmetric, transverse, and horizontal modes. Likely sources of these oscillation modes in these earlier wind tunnel experiments are turbulence and low frequency pulsations in airspeed.

Raindrop axis ratios can differ in other ways from the equilibrium shapes measured by Pruppacher and Beard. One recognized cause is the canting of raindrops in wind shear near the ground (Brussard, 1974) and by severe turbulence in thunderstorms (Beard and Jameson, 1983). However, such tilting is caused by extreme conditions that would not



Fig. 1. Shape of large drops falling at terminal velocity having equivalent spherical diameters of  $D = 8.00, 7.35, 5.80, 5.30, 3.45$ , and  $2.70$  mm.

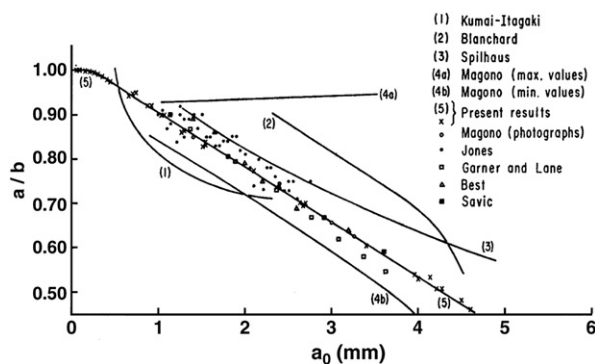


Fig. 2. Axis ratio determined from photographs of water drops suspended in the UCLA wind tunnel for sizes up to about 4.5 mm radius (9 mm diameter).

normally occur in rain showers except just above the ground during strong winds (Saunders, 1971).

The fact that raindrop oscillate is readily apparent from the broken fall streaks observed in backscattered light against a dark background. This is easily seen in rain at night if the observer has a spot light above his head to illuminate the rain against a dark background. The effect is so evident that there is available software to add the dashed streaks of simulated rain to movie scenes (e.g., see Garg and Nayer, 2007). This phenomenon occurs when light is internally reflected near the angle of the primary rainbow (Volz, 1960). Such observations in nocturnal rain reveal that the onset of natural raindrop oscillations occurs at  $D = 1.0$  mm (Beard and Tokay, 1991) and extends to the largest size that has been measured,  $D = 3.0$  mm (Tokay and Beard, 1996). Calculations in these two papers demonstrate that external forcing of oscillations by collisions, wind shear and turbulence are too weak to maintain the oscillations against viscous dissipation. The authors believed that the likely sources of oscillations include (i) positive feedback as the drag force varies with axis ratio (Beard, 1984), and (ii) pressure perturbations produced by transverse eddy shedding, as well as resonance between the eddy shedding frequency and oscillation frequency.

Small amplitude oscillations of a spherical drop occur in  $m = 1 + n$  physically distinct spherical harmonic modes (Pruppacher and Klett, 1997). There are three distinct modes at the fundamental frequency that can be expected to distort the axisymmetric equilibrium shape: axisymmetric, transverse and horizontal modes. Oscillations in the transverse mode result in oscillations in a vertical plane where the shape perturbation extends the axes both vertically and horizontally one side and then the other side. In the plane of the oscillation the axis ratio varies like a rectified sine wave.

The physically distinct oscillation modes for a raindrop may excite different types of eddy shedding. For example, the oscillations in the axisymmetric mode ( $m = 0$ ) would tend to trigger axisymmetric vortex rings, whereas at  $m = 1$ , the transverse mode oscillation would favor eddy shedding from alternate sides of the drop. This latter type of eddy shedding was observed by Saylor and Jones (2005) in their visualized wake study of water drops suspended in a vertical air stream of a wind tunnel and was shown to be correlated with transverse mode oscillations.

Raindrop oscillation shapes are shown in Fig. 3 for a sequence of “multimode” oscillations composed of the

axisymmetric ( $m = 0$ ) and transverse ( $m = 1$ ) modes of the fundamental frequency. These theoretical shapes in the mixed mode sequence bear a strong resemblance to photographic sequences of oscillating drops obtained in a vertical wind tunnel by Musgrove and Brook (1975).

Collisions between raindrops are another source of distortion. For dissimilar size drops, such as a raindrop of  $D = 2$  mm raindrop and a drizzle of  $d = 0.2$  mm, the impact results in a coalesce without greatly disturbing the raindrop shape. (In this case the mass of the raindrop is 1000 times the mass of the drizzle drop). The process of accretion of cloud drops is responsible for rain formation in warm-season clouds such a trade-wind showers.

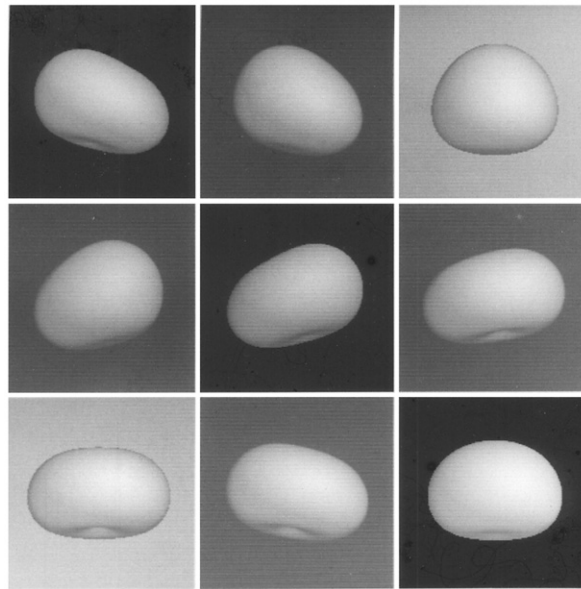
Most information on collisions *between* raindrops has been obtained from laboratory studies. In the experiments of McTaggart-Cowan and List (1975), breakup was evaluated for diameters of 3.0 to 4.8 mm as they collided with smaller drops of 1.0 to 1.8 mm. Out of over 25,000 attempts, 712 collisions were used to classify three predominate breakup types, characterized as neck (27%), sheet (55%) and disk (18%), as shown in Fig. 4. Disk collisions are a major cause for the depletion of large raindrops and may limit the size of raindrops. The more common types of breakup (neck, sheet) result in a large drop fragment with transient raindrop axis ratios that can exceed 1.0.

In contrast to this breakup, extremely large raindrops can grow by accretion of cloud drops in the relative absence of smaller raindrops. The convergence of trade winds and the morning drainage flow off of Mauna Loa east of Hilo, Hawaii combined to produce the right ingredient for raindrops as large as  $D = 8$  mm (Beard et al., 1986). This special circumstance results in a sufficiently strong and steady updraft to suspend a few raindrops while they grow by accretion of cloud drops in the absence of smaller raindrops (Raubert et al., 1991). The photograph of largest drop the UCLA wind tunnel study was also about 8 mm diameter and its axis ratio was 0.53 (see Figs. 1 and 2,  $a_0 = 4$  mm).

### 3. Theory

Raindrop oscillations were modeled by Beard (1984) assuming potential flow for ellipsoidal variations driven by changes in surface energy and gravitational potential energy with linear dissipation of kinetic energy. The oscillations were calculated in a local reference frame moving at the terminal velocity of the raindrop. The equilibrium axis ratio was found from the minimum in the energy conservation equation that includes the surface and gravitation potential energy with respect to the reference frame moving at a constant terminal speed. Since the surface area for an oscillating raindrop is a minimum when it is spherical, the time-average axis ratio for a pure axisymmetric oscillation is more spherical, than its equilibrium axis ratio. These oscillation model result of Beard (1984) are quantitatively similar to a Navier–Stokes fluid evaluated by Foote (1973) for axisymmetric oscillations of a water drop in the absence of gravity.

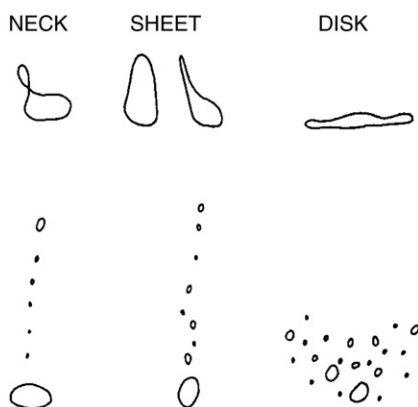
Fig. 5 shows the variation of the oscillation energy with axis ratio for an axisymmetric oscillation of a  $D = 3$  mm raindrop. The asymmetry results from a sum of the surface energy and gravitational potential energy on a reference frame falling at terminal velocity. The former is a minimum



**Fig. 3.** Oscillation sequence for a 5 mm diameter raindrop in the eight picture from top left to bottom center (lower right square depicts the static shape). The oscillation is a mixture of the two modes of the fundamental harmonic, 90° out of phase, with amplitudes of 0.75 mm. The white squares show the pure axisymmetric mode, whereas the black ones are the pure transverse mode. The grey squares are equal mixtures. Note that the view is from slightly below horizontal to show the base of the drops which have a dimple.

for a sphere and latter is a minimum at the smallest axis ratio. The diagonal line is the time-average axis ratio as a function of oscillation energy. The probability density functions are similar to secant curves with minimum probabilities at the average axis ratio  $\langle\alpha_a\rangle$  for the axisymmetric oscillations. This theory, which was also applied to the horizontal mode, provided guidance for evaluating oscillation modes from axis ratio data.

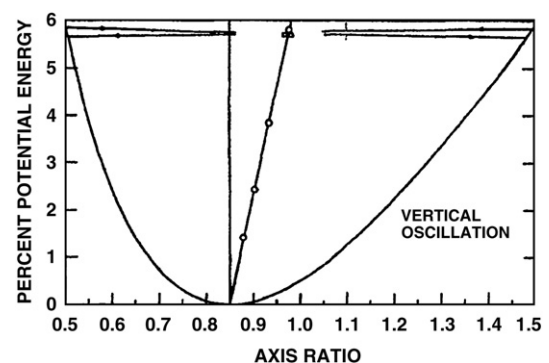
Formulas for the potential flow oscillations were developed with good accuracy for the time average axis ratio as simple functions of the minimum ( $\alpha'$ ), maximum ( $\alpha''$ ) or equilibrium ( $\alpha_0$ ) axis ratios for the axisymmetric modes. The time average for the axisymmetric mode can be calculated using  $\langle\alpha_t\rangle = 0.5(\alpha' + \alpha'') \sim 0.5(\alpha' + \alpha_0^2/\alpha')$ .



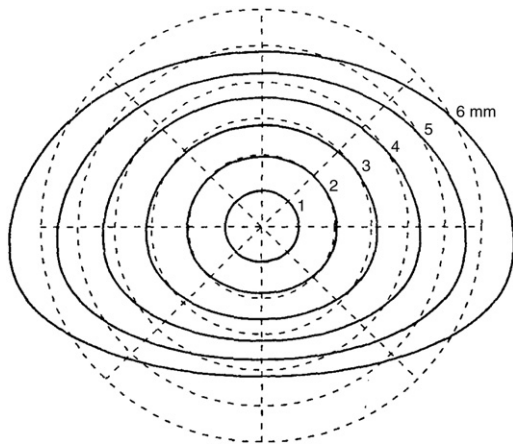
**Fig. 4.** Schematic illustration of the three most common types of drop breakup (MacTaggart-Cowan and List 1975).

Raindrop axis ratios have also been evaluated using numerical methods by Beard and Chuang, 1987). The force balance at the surface was determined using Laplace's equation with an internal hydrostatic pressure and an external aerodynamic pressure distribution based on spheroidal drop distortions. The drop shape was calculated by integration from the upper pole with the initial curvature determined by iteration on the drop volume. The shape was closed at the lower pole by adjusting pressure drag and the effective drop weight to achieve an overall force balance. The numerical shapes are shown in Fig. 6.

The numerical shape is also shown in Fig. 7 for comparison with the shapes for sessile drop (flat base) and the Pruppacher and Pitter perturbation model (indented base). The model of Beard and Chuang does not predict an indentation of the base for the  $D = 5$  mm raindrop.



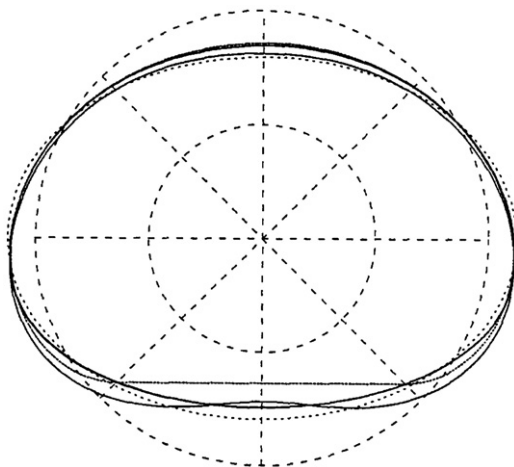
**Fig. 5.** Potential energy (%) for the axisymmetric mode as a function of axis ratio for a  $D = 3$  mm raindrop. The slanted vertical line is the average axis ratio for the oscillating.



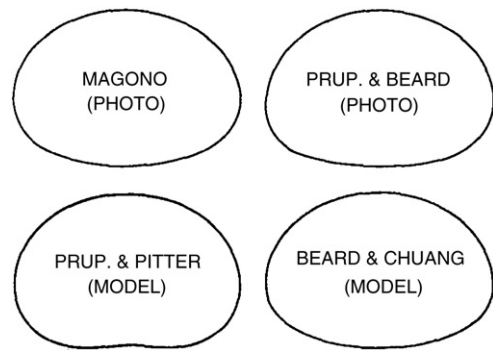
**Fig. 6.** Computed shapes for  $D=1, 2, 3, 4, 5$  and  $6$  mm with origin at the center of mass. Shown for comparison are dashed circles of diameter  $D$  divided in  $45^\circ$  sectors.

A further comparison of  $5$  mm drop shapes is shown in Fig. 8 for shapes obtained from the wind tunnel observations of Magono (1954) as well as Pruppacher and Beard and the model of Pruppacher and Pitter. Again there is no evidence from observations of an indentation at the base of a  $5$  mm drop.

In contrast to earlier models for an oblate spheroid response to gravity or the perturbation response to the aerodynamic pressure for a sphere (Imai, Savic, Pruppacher and Pitter), the model of Beard and Chuang provides the large amplitude response to both the hydrostatic and aerodynamic pressures modified for distortion. The model results also provided a bound on the axis ratio of raindrops with an uncertainty of about  $1\%$  and in very good agreement with the



**Fig. 7.** Comparison of cross sections for a  $D=5$  mm water drop. The numerical result for a sessile drop is distinguished by a flat base (small dashes), whereas the numerical model results for the mean forcing from aerodynamic and hydrostatic pressures have a rounded base (solid line). The shape with the recurved base is the perturbation model of Pruppacher and Pitter (lighter solid line). Also presented is an ellipse of axis ratio  $0.71$  (medium dashes) and circles (large dashes) scaled to  $2.3$  and  $5$  mm diameter with  $45^\circ$  sectors.



**Fig. 8.** Comparison of drop profiles for  $D=5$  mm with model results.

wind tunnel measurements of Pruppacher and Beard (1970), even for drops as large as  $D=8$  mm.

The axis ratio variation with diameter from the numerical model is compared in Fig. 9 with previous values from measurement and theory. The measurements were mostly obtained from photographs of water drops suspended in vertical wind tunnels. Although care was used in obtaining representative axis ratios, there is scatter from transient distortions excited by various means used to stabilize the drops.

The wind tunnel measurements of Pruppacher and Beard are within the bounds of the numerical model for  $D>3$  mm whereas the measurements of Pruppacher and Pitter appear to be closer for small drops. The axis ratio data from Clift et al. (1978) is a fit to experimental data using dimensionless groups. Their general approach captured the magnitude and trend in axis ratios for raindrops.

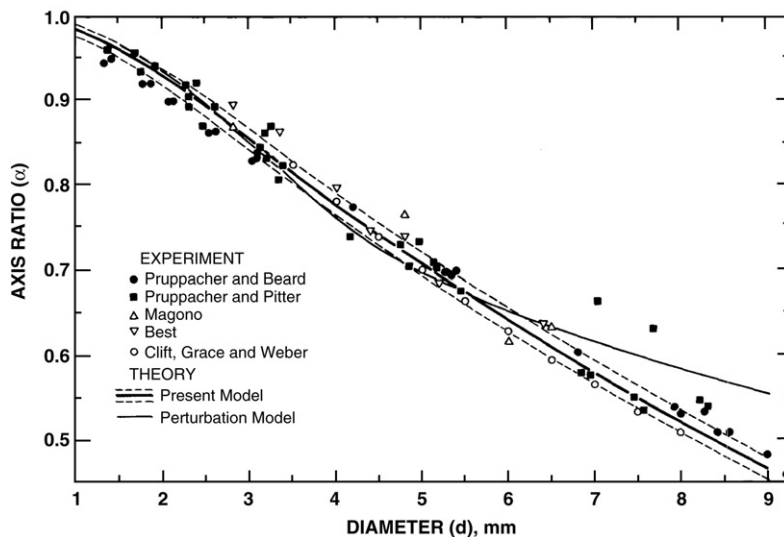
The axis ratio variation from the numerical model can be represented by a 4th order polynomial from Andsager et al. (1999):

$$\alpha_N = 1.0048 + 0.0057D - 2.628D^2 + 3.682D^3 - 1.677D^4 \quad (2)$$

This formula is preferred to the one in Pruppacher and Klett (1997) which is based on electrified drop model of Chuang and Beard (1990), because more accurate terminal velocities could be used in the absence of electrical forces.

Raindrop shapes influenced by electric fields and charges been evaluated by Beard et al. (1989a) using a set of first-order perturbation equations with spherical harmonics to help reconcile the apparent inconsistencies with the cosine series for raindrop shape originated by Savic (1953) and modified by Pruppacher and Pitter (1971b). The shape coefficients were solved from nine equations with nine shape coefficients using an appropriate pressure balance equation. The resulting raindrop shapes without electrical stresses are shown in Fig. 10. The axis ratios from the perturbation model of Beard et al. (1989b) using the consistent first-order perturbation equations with spherical harmonics are nearly the same as results from the model of Pruppacher and Pitter. All models except Green's hydrostatic model predict similar axis ratios for small raindrops.

The electric stress was included as a first-order perturbation assuming a conducting sphere in a uniform vertical field. The effect of a net charge was approximated by the Coulomb-force reduction of surface tension for a spherical drop. These



**Fig. 9.** Experimental and modeling results for axis ratio of raindrops as a function of drop diameter. The dashed lines are the upper and lower bound estimates from the numerical model of Beard and Chuang.

perturbation drop shapes for electric distortion are altered by coupling between shape and fall speed. A simple axis ratio formula was developed for small perturbations based on the principal shape coefficient for the distortion from the aerodynamic pressure as well as the electric field and charge. This equation, when coupled with a shape-drag relation yielded a cubic equation for axis ratio (see Table 5 therein).

Calculated axis ratios from the Beard et al. (1989a) perturbation model are consistent with wind tunnel observations for non-oscillating drops in vertical electric fields, and generally agreed with the numerical findings of Chuang and Beard (1990) for electrostatic–aerodynamic shapes. The results indicated detectable shape changes in thunderstorms – produced directly by strong electric fields, but modified considerably by changes in fall speed, with and without charge.

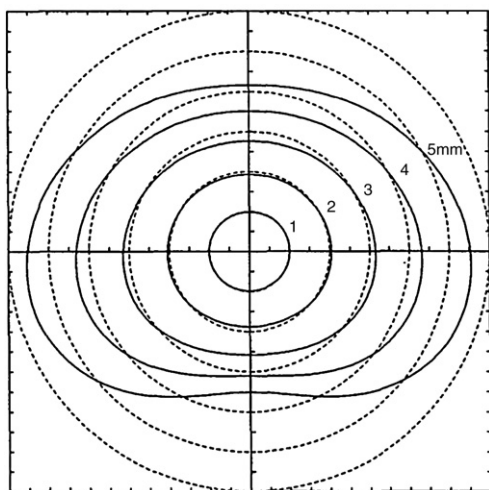
The largest effect would be produced by the maximum vertical electric field that occurs just before the initiation of lightning where the vertical stress would produce nearly spherical raindrops producing large changes in  $Z_{DR}$ .

In a detailed study of oscillation frequencies, Feng and Beard (1991) used the method of multiple-parameter perturbations to evaluate the oscillation characteristics of raindrops. The equilibrium drop shape was approximated as an oblate spheroid in response to the symmetric pressure distribution of potential flow. The resulting oscillation frequencies for  $m = 0, 1$ , and  $2$  are plotted in Fig. 11 for comparison with the Rayleigh frequency (dashed line) and wind tunnel measurements. The cinema data include observations from movie frames of oscillating drops by Blanchard (1950) and Brook and Latham (1968) and also the microwave frequency measurements of Brook and Latham, as well as Nelson and Gokale (1972).

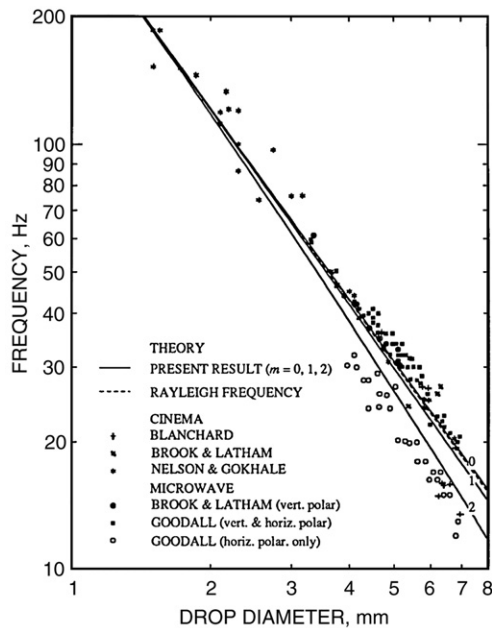
As evident from Fig. 11, the oscillation frequencies split from the Rayleigh frequency with a slight increase for axisymmetric mode ( $m = 0$ ), a slight decrease for transverse mode ( $m = 1$ ), as well as a more significant decrease for the horizontal mode ( $m = 2$ ). These trends were previously predicted by the theoretical model of Beard (1984).

Note that the data in Fig. 11 were obtained in wind tunnels with open test sections and much higher turbulence levels than Pruppacher's wind tunnels. The drops may be oscillating in more than one mode as depicted in Fig. 5.

In Fig. 11 a data set was obtained by Goodall (1976) by detecting scattering of horizontally and vertically polarized microwaves. Two frequencies were clearly distinguished. The higher one was detected at both polarizations that indicates variations in both the horizontal and vertical axes. The lower frequency was detected for only for horizontal polarization caused by variations in only the horizontal axes. As discussed in the next section for the horizontal mode ( $m = 2$ ), the horizontal axis varies while the vertical axis remains fixed. This “horiz. polar. only” is a clear indication of the frequency of horizontal mode oscillations. The other two polarization



**Fig. 10.** Perturbation aerodynamic shapes for  $d = 1, 2, 3, 4$ , and  $5$  mm with origin at the center of mass. Shown for comparison are dashed circle of  $d = 1$  to  $6$  mm.



**Fig. 11.** Frequencies of the three fundamental mode oscillations for two-lobed modes. Data points are from observations. Solid lines were obtained from theory with curves 0, 1, and 2 denoting the  $m = 0, 1$ , and 2. The dashed line is Rayleigh's result for the two-lobed mode.

measurements could be caused by either the axisymmetric or transverse modes because both the vertical and horizontal axes vary in these modes. The differences in frequencies between perturbation theory and the wind tunnel data may indicate the trends in frequency for finite amplitude oscillations.

#### 4. Laboratory studies of drop oscillations

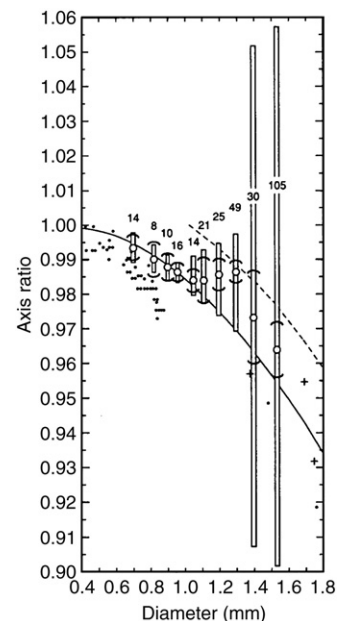
Experiments have documented that water drops of 1 mm diameter and larger falling in still air continually oscillate up to the largest sizes studied (Beard et al., 1989b, Beard et al., 1991, Beard and Kubesh, 1991, Kubesh and Beard, 1993, Andsager et al. (1999)). The smallest raindrops that oscillate have a Reynolds number equal to the critical value for the onset of eddy shedding from a sphere. The extension to larger raindrops may be a consequence of a lock-in resonance whereby eddy detachment is triggered by the natural oscillations of the raindrop. This type of resonance has been observed for a cylinder in cross flow where the eddy shedding is triggered at the natural vibration frequency of the cylinder rather than its critical Reynolds number for an unstable wake (Williamson and Roshko, 1988). [Note that a lock-in resonance may favor axisymmetric oscillations for eddy shedding of vortex rings or transverse mode oscillations if eddies detach from alternate sides of the fall axis.]

The initial studies by Beard et al., 1989a, Beard et al., 1991) and Beard and Kubesh (1991), were focused on drops near 1 mm diameter where the wake becomes unsteady and the eddy shedding frequency is close to the oscillation frequency (Gunn, 1949). In these axis ratio measurements a camera with magnification was used along with glass sphere calibrations to obtain a significantly higher axis ratio resolution than in the earlier studies of Pruppacher and

Beard (1970) and Pruppacher and Pitter (1971a). In addition, the drop sizes could be more accurately calculated using the measurements of drop generator frequency and its flow rate. The results in Fig. 12 show the mean axis ratio, experimental uncertainty and 95% confidence interval.

The data from this experiment illustrate the following axis ratio behavior for small raindrops:

- (i). Drops with diameters less than 1 mm have equilibrium axis ratios similar to the measurements of Pruppacher and Beard and in agreement with the equilibrium axis ratios determined by the numerical model of Beard and Chuang (1987). The small scatter in axis ratios is consistent with experimental errors.
- (ii). Drops from 1.0 to 1.3 mm diameter oscillate between the equilibrium axis ratio and a slightly larger axis ratio. This behavior is consistent with an azimuthal average of transverse mode oscillations at fundamental frequency ( $n = 2, m = 1$ ). The one-millimeter drop size has a Reynolds number that matches the onset of eddy shedding for a sphere. The eddy shedding from alternate sides of the fall axis has a pattern that matches the transverse mode oscillations.
- (iii). The two drop sizes of 1.40 and 1.54 mm diameter appear to oscillate in an axisymmetric mode ( $n = 2, m = 0$ ). The greater oscillation amplitude may indicate a lock-in resonance with eddy shedding. Note that the shift in the average axis ratio is of order 0.01 and is much less than the offsets of 0.05 observed by Jones (1959) for larger raindrops of  $D = 2$  to 4 mm.



**Fig. 12.** Axis ratio of small water drops falling at terminal velocity measured by Beard et al. (1989a, 1989b, 1991) and Beard and Kubesh (1991). The solid curve is the axis ratio from the numerical model of Beard and Chuang (1987) and the dashed curve shows the postulated shift in axis ratio based on polarization radar and disdrometer measurement in rain (Goddard and Cherry, 1984).

In the above studies, the axis ratios and fall behavior for water drops of  $D = 1.1$  to  $1.5$  mm were found to be consistent with oscillations and horizontal drift in response to various Reynolds-number dependent wake configurations as determined from observation of freely falling water drops in organic liquids. The small scattering of axis-ratio above equilibrium and the maximum drift corresponded to the onset of an asymmetric pulsating wake found by Magarvey and Bishop (1961a, Magarvey and Bishop, 1961b) whereas the large axis ratio scatter above and below equilibrium was an apparent axisymmetric oscillation response.

By using light refracted between the angles of the primary and secondary rainbows in oscillating drops, Beard and Kubesh (1991) determined that two oscillation frequencies were present for all drop sizes. The significant oscillation modes for smaller drops ( $d = 1.04$ – $1.30$  mm) were the transverse modes of the fundamental and first harmonic, whereas significant oscillation modes for larger drops ( $1.40$ – $1.54$  mm) were the axisymmetric mode of the fundamental and the transverse mode of the first harmonic. The study shows that small raindrops oscillate as a response to eddy shedding. It was noted that these postulated oscillation modes would produce the trends in axis ratios inferred for small raindrops from the field studies of Goddard and Cherry (1984) and Chandrasekar et al. (1988).

Although the oscillation modes could not be directly evaluated from the measurements shown in Fig. 12, the probable oscillation modes were identified by evaluating the oscillation amplitudes for the normal modes as a function of distortion energy for both positive and negative changes from equilibrium axis ratios:  $+\Delta\alpha = \alpha_{\max} - \alpha_0(D)$  and  $-\Delta\alpha = \alpha_0(D) - \alpha_{\min}$ . Since the surface area of a spherical harmonic mode is just the area of a sphere plus the area of the distortion, the distortion energy ( $\Delta E$ ) for a raindrop was computed from the increase in surface area from the equilibrium axis ratio.

The distortion energy as a function of axis ratio amplitude is plotted in Fig. 13 for the oscillation modes at the two lowest frequencies. The axis ratio amplitudes were estimated from the extremes in Fig. 12, using the mode orientation ( $\phi$ ) that produced the maximum change in axis ratio. In this figure only one energy is needed for modes having only positive or negative axis ratio change, i.e., mode (2,1) at the fundamental and all of the first harmonic modes.

The amount of energy needed to produce comparable changes in axis ratio varies considerably with the oscillation mode. The axisymmetric mode (2,0) produces the largest  $\Delta\alpha$  for a given energy, because the vertical and horizontal axes oscillate in opposition and the axis ratio is independent of azimuthal angle. In all other modes there are factors such as compensating changes that reduce the magnitude of  $\Delta\alpha$  compared to mode (2,0) at the same oscillation energy.

In Fig. 14 the distortion energies are plotted for the range in axis ratios shown in Fig. 12 using formulas give in Tables A1 and A2 in Beard and Kubesh (1991). The axisymmetric oscillations occur at 0.1% to 0.3% of the surface energy, considerably less energy than the transverse mode oscillation for the smaller sizes. If transverse mode oscillations were occurring for  $d = 1.40$  or  $1.54$  mm, the axis ratio behavior would be lost in the larger amplitude of the axisymmetric mode. Since there are no axis ratios below the equilibrium for

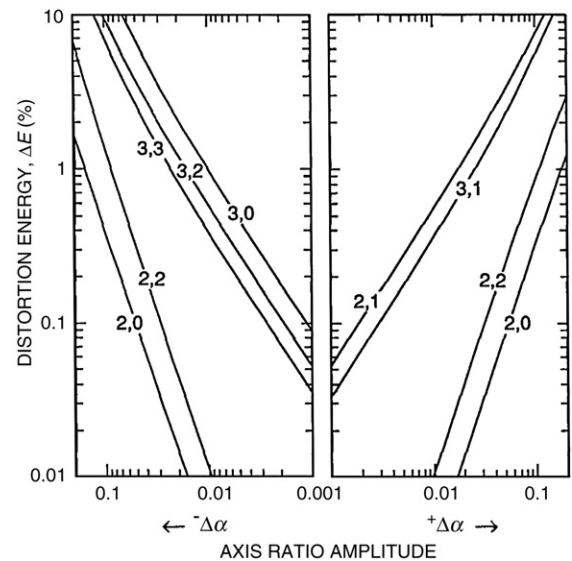


Fig. 13. Distortion energy as a function of the axis ratio amplitude expressed as a percent of the surface energy of an equivalent volume sphere [ $+\Delta\alpha = \alpha_{\max} - \alpha_0(D)$ ,  $-\Delta\alpha = \alpha_0(D) - \alpha_{\min}$ ] assuming the particular view having the maximum  $\Delta\alpha$ .

transverse modes, the uncertainty, was estimated using the largest magnitude of (2,1).

The oscillations for the smaller drop sizes show a steady increase in distortion energy for the transverse modes, as a possible consequence of resonance, e.g., from the transverse pressure fluctuations produce by eddy shedding. Both harmonics need to be present to account for the observed frequencies of  $n = 2$  and  $3$ .

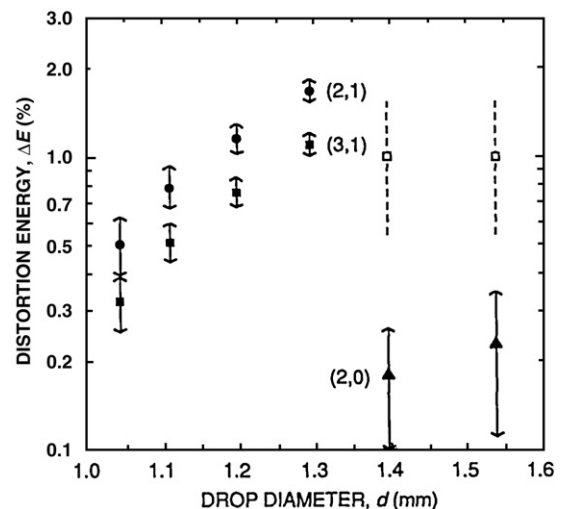


Fig. 14. Distortion energy for the drop sizes plotted Fig. 12 expressed as a percent of the surface energy of an equivalent volume sphere. Three modes are shown: (2,1) and (3,1) for smaller drops (corresponding the observed one-sided scatter in axis ratio) and (2,0) for larger drops corresponding to the two-sided scatter in axis ratio). The possibility of mode (3,1) for larger drops, around an energy of 1%, is suggested by the open squares with dashed error bars.

Major conclusions from the Beard–Kubesh laboratory study of the oscillations of freely falling drops of  $D = 1.04$  to  $1.54$  mm:

1. Information obtained by backscattered light established that two frequencies were present for all drop sizes.
2. Significant oscillation modes for large sizes drops of  $D = 1.4$  and  $1.54$  mm) are the axisymmetric mode of the fundamental and the transverse mode of the first harmonic.
3. Primary resonance is probably responsible for the transverse modes because of the match in frequencies between the forcing and response, and because the spatial pattern of eddy shedding should force these modes.
4. Secondary resonance would account for the axisymmetric mode in larger drops. This mode is a subharmonic of the forcing frequency and there is no requirement for the forcing pattern to match the response. Thus axisymmetric oscillations appear to be a natural response to eddy shedding for all larger raindrop sizes.

In summary, the study of Beard and Kubesh revealed that small raindrops oscillate as a response to eddy shedding. They postulated that the oscillation modes discovered in their laboratory experiments would also produce the trends in axis ratio inferred for small raindrops from field studies (Goddard and Cherry; Chandrasekar et al.). Secondary resonance was also postulated as the cause of raindrop oscillations detected in field studies for much larger raindrops.

Two subsequent sets of axis ratio measurements for larger drops were made using a 25 m fall column installed in a stairwell located at the University of Illinois at Urbana-Champaign. In the first study Kubesh and Beard (1993) found that the average axis ratios for water drops of  $D = 2.0$  and  $2.5$  mm were slightly higher than the equilibrium values as were the axis ratios for  $D = 1.4$  and  $1.5$  mm in the earlier study (see Fig. 15a). This behavior is consistent with axisymmetric oscillations (see also Fig. 5).

The results of a subsequent study in the same fall column by Andsager et al. (1999) are shown in Fig. 15b. The average axis ratios for  $D = 2.5$  and  $3.6$  mm were found to be close the

equilibrium values, whereas the axis ratios for  $D = 2.9$  and  $4.0$  mm were slightly higher than equilibrium values. Both the 1993 and 1999 experiments obtained axis ratios for  $D = 2.5$  mm, and the mean axis ratios differed slightly: 0.92 versus 0.89. Note that the vertical scale in Fig. 15a is expanded by a factor of 2 compared to Fig. 15, so the offsets of axis ratios from the non-oscillating equilibrium values for these two sets of measurements are similar.

A new hypothesis is that oscillations induced by the drop generators may have biased the measurements. In the Kubesh and Beard experiments, transverse mode oscillations were observed just below the drop generator which was the same type used in earlier studies for smaller drops of Beard et al (1991). Since the average axis ratio of the transverse mode is higher than the equilibrium axis ratio, if this mode persisted in the 20 m fall column, it would produce an average axis ratio that is higher than the equilibrium value.

In the Andsager et al. experiment, axisymmetric oscillations were observed as drops were released from a needle at a constant drip rate. The average axis ratio for axisymmetric oscillations is only slightly higher than the equilibrium values for small oscillation amplitude, e.g., as obtained from Fig. 15 for  $D = 3$  mm using an amplitude of order 0.01.

The overall conclusion from the laboratory measurements of axis ratios shown in Figs. 12 and 15a and b is that raindrops from  $D = 1.4$  to  $4.0$  mm naturally oscillate with time-average axis ratios close to equilibrium or slightly higher. There also appears to be transverse mode oscillations for small raindrops of  $D = 1.1$  to  $1.3$  mm apparent from the positive offset of the average axis from the equilibrium values in Fig. 12.

## 5. New axis ratio measurements

In the recent laboratory study by Szkal et al. (2008) the axis ratios of water drops as large as  $D = 7$  mm were measured using the wind tunnel in Mainz, Germany built under the supervision of Hans Pruppacher at Johannes Gutenberg University. The axis ratio for drops up to about  $D = 7$  mm were measured, but those between  $0.8$  and  $1.0$  mm drifted horizontally and therefore could not be stably supported.

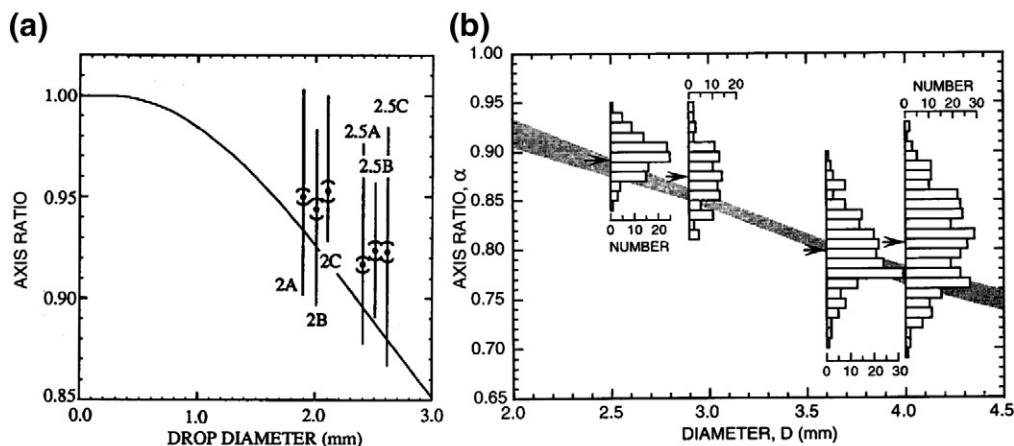
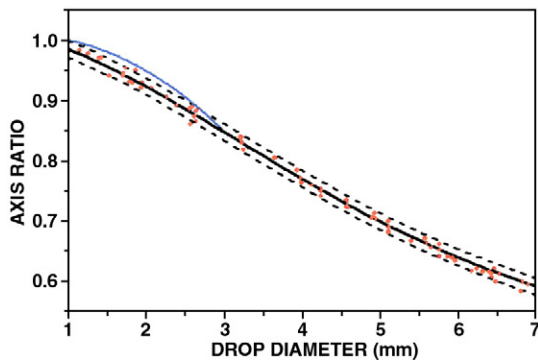


Fig. 15. a. Axis ratio for  $D = 2.0$  and  $2.5$  mm drops falling at terminal velocity (Kubesh and Beard, 1993). Vertical bars show the range of measured values. b. Axis ratio for  $D = 2.5, 3.0, 3.6$  and  $4.0$  mm drops falling at terminal velocity (Andsager et al., 1999). Shaded line depicts the uncertainty in equilibrium axis ratio.



**Fig. 16.** Time average axis ratio of raindrops falling at their terminal velocities measured in the Mainz vertical wind tunnel (red symbols). The equilibrium axis ratio from the model of Beard and Chuang (1987) is plotted by solid, the upper and lower limit of the model with dashed line. The radar-disdrometer derived axis ratios postulated by Goddard and Cherry (1984) are plotted by blue line.

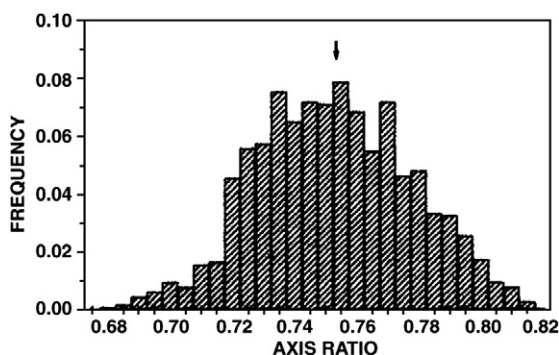
This finding is consistent with the lateral drift measured by Beard et al (1991) and attributed to an asymmetric wake.

Water drops from  $D = 1.1$  to  $7.0$  mm were observed to oscillate with mean axis ratios that decrease from about  $\alpha = 0.99$  at  $D = 1.1$  mm to  $\alpha = 0.6$  at  $D = 7$  mm. The oscillation amplitudes increased from  $\pm 0.02$  at  $1$  mm to  $\pm 0.15$  at  $7$  mm. The findings from this study are within the bounds of the numerical model of Beard and Chuang (1987) as shown in Fig. 16.

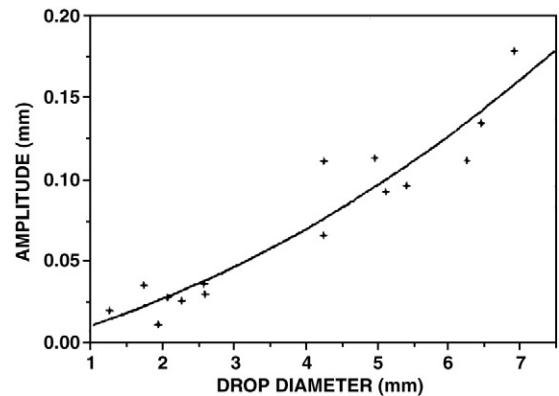
An axis ratio distribution from this study for a  $4.24$  mm diameter drop is shown in Fig. 17 with the time average axis ratio indicated by an arrow. This nearly normal distribution is remarkable similar to the Fig. 15 distribution from Andsager et al (1999) for  $4$  mm drops.

Larger drops had larger oscillation amplitudes as shown in Fig. 18. The authors suggest that this may be a consequence of vortex shedding since the vortex energy increases with the drop size.

The fact that raindrop oscillations appear to be mostly axisymmetric is consistent with the results of Andsager et al for drops of  $D = 2.5$  to  $4.0$  mm. An increasing oscillation amplitude (energy) with drop size is also evident from laboratory results for  $D = 1$  mm to  $4$  mm (Figs. 12 and 15).



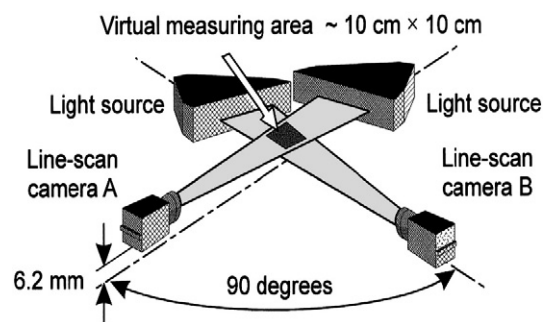
**Fig. 17.** Histogram of an axis ratio for a  $4.24$  mm diameter drop. The time average axis ratio value is indicated by an arrow.



**Fig. 18.** Axis ratio amplitude for drop oscillations. The best fit is a second order polynomial.

In the recent paper by Thurai et al (2009a) the axis ratios of water drops from  $D < 1$  mm out to  $7$  mm obtained by Szkall et al. (2008) were compared to axis ratios obtained for water drops generated from a fire hose off an  $80$  m bridge in Carinthia, Austria (Thurai and Bringi, 2005). After about  $5$  m of fall the stream from the bridge separated into a much wider column of drops that simulated an artificial rain (the distance for a drop to reach terminal velocity is only about  $20$  m). The remaining  $60$  m of fall allowed the drop stream to widen and individual drops to achieve their terminal velocity. On the ground beneath the bridge the vertical and horizontal axes of drops were measured using a high-resolution 2D-video disdrometer (Schönhuber et al, 2008).

The 2D-video disdrometer has two, orthogonally placed line scan cameras (line scan frequency of  $55$  kHz) and illumination sources which give orthogonal 'views' of the raindrop as it passes through the  $10 \times 10$  cm sensor area (Fig. 19). The precisely calibrated vertical distance between the two light planes enables the accurate measurement of fall speed. From the two views, the drop volume (or, equivalent spherical diameter  $D$  or  $D_{eq}$ ), the axis ratio and smoothed contours are obtained. The instrument measures the rain rate as the volume flux of drops through the  $10 \times 10$  cm area per unit time. Since the fall speed is known for each drop, the



**Fig. 19.** Schematic of the 2D-video disdrometer showing the orthogonally placed line scan cameras and the illumination sources. There are two light planes separated by very accurately calibrated vertical (height) distance of (nominally)  $6$  mm. From Kruger and Krajewski (2002).

drop size distribution,  $N(D)$ , is also obtained. Further data processing of the two smoothed contours also gives the angle between the drop's symmetry axis with respect to the local vertical direction (also termed canting angle in the radio-wave propagation community; two estimates are available, one from each view). From the two canting angles, the polar and azimuthal angles can also be derived. For rain rate, fall speed and  $N(D)$  determination, the smallest diameter is close to 0.35 mm, while for axis ratio, smoothed contours and canting angle determination are possible for  $D > 1.5$  mm but with lower accuracy for smaller drops due to quantization effects (horizontal and vertical resolutions are approximately 0.2 mm).

As seen in Fig. 20 (from Thurai et al, 2009b), shown below, the measured variance in axis ratio for the bridge experiment ( $\pm \sigma$  vertical bars) increases with drop size from  $D = 1.5$  to 4.5 mm and remains relatively large out to  $D = 9$  mm, the largest size category measured. Instrumental effects such as the finite resolution of around 0.2 mm are the likely source of the axis ratio dispersion for drops smaller than  $D = 1.5$  mm.

In the Mainz wind tunnel, drops smaller than  $D = 2$  mm oscillated with an amplitude of about  $\Delta AR = \pm 0.03$  (Fig. 18). For larger sizes the amplitude gradually increased to  $\pm 0.07$  at  $D = 4$  mm and continued to increase out to the largest size of  $D = 7$  mm. Drops between about 0.8 and 1.2 mm could not be stably supported in the wind tunnel, apparently because of the sideways drift produced by an asymmetric vortex trail (Beard et al., 1991).

In the bridge experiment the finite 2DVD resolution increases the standard deviations for diameters less than about 1.5 mm. The standard deviation is a minimum at about  $D = 2$  mm and gradually increases back to  $\pm 0.05$  at  $D = 3.5$  mm. For drops larger than about  $D = 4$  mm the standard

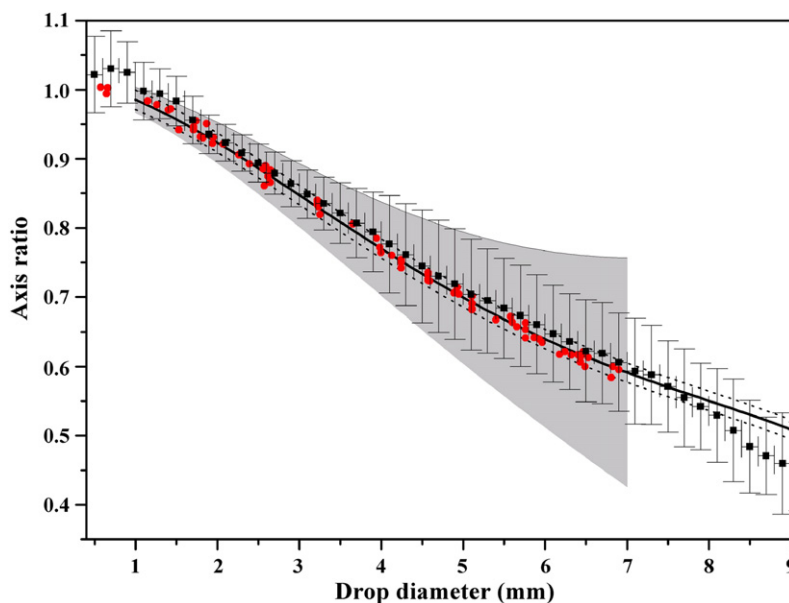
deviations are  $\pm 0.06$  to  $\pm 0.09$ . In both experiments the average axis ratios for drops larger than  $D = 1.5$  mm are close to values from the numerical model axis ratios of Beard and Chuang and within the model boundaries.

## 6. Conclusions from recent axis ratio experiments

The comparison in Thurai et al (2009b) of the wind tunnel measurements of Szkal et al. (2008) and the data from the bridge experiment of Thurai and Bringi (2005) provide a revised picture for the axis ratios of raindrops larger than about 1.4 mm diameter: the average axis ratio are within the bounds of the theory of Beard and Chuang (1987) for the equilibrium axis ratio of raindrops.

One major difference between the earlier laboratory studies and the recent wind tunnel-bridge data comparison is the time for the generated drops to achieve their equilibrium axis ratios. The fall columns of 20 m used in the laboratory studies of Beard and Kubesh and Andsager et al. allowed only about 10 s between drop generation and measurement. In the bridge experiment there were at least 40 s between drop generation and the shape measurements by the 2DVD. The wind tunnel measurement of axis ratios did not have a short time restriction.

A possible reason for different results is that there was insufficient time in the laboratory experiments for the transient shapes produced by the drop generators to be replaced by equilibrium shapes. Certainly the drops in 80 m bridge experiment and in the wind tunnel measurements in Mainz have much longer times to equilibrate, and therefore should provide axis ratios that are closer to raindrops falling in air. The UCLA wind tunnel with its low turbulence and



**Fig. 20.** Mean axis ratio vs drop diameter from wind tunnel experiment (red dots) and the bridge experiment 2DVD measurements (black dots). The shaded area represents the drop oscillations from the wind-tunnel data, and the vertical bars ( $\pm 1\sigma$ ) represent the standard deviation of the 2DVD axis ratio distributions. The small horizontal bars represent the bin width of the 2DVD interval. The solid line is the Beard–Chuang model axis ratio and the dotted lines are the upper and lower bounds.

uniform velocity also provided a steady free fall environment and a reliable estimate of raindrop shape.

## 7. Drop shapes and canting angles from 2D-video disdrometer

### 7.1. Shapes from the 80 m fall (artificial rain) experiment

The 80 m fall experiment using the 2D-video disdrometer (2DVD) referred to in the previous section was conducted outdoors during calm wind conditions. The experiment enabled the contoured shapes of more than 115,000 drops to be determined for drops as large as 9 mm equivalent drop diameter,  $D_{eq}$  (see [Thurai et al., 2007](#)). The processed images for 3, 4, 5 and 6 mm drops are given in [Fig. 21](#). The frequency of occurrence is contoured on a log(probability) scale, in order to highlight the mean shapes as well as the shape variations. The individually processed images were subsequently fitted to a conical-like (henceforth simply referred to as the conical fit) equation given in Eq. (1) of [Thurai et al. \(2007\)](#) to represent the mean shape in terms of  $D_{eq}$ . The conical equation with three parameters (modified version of that given in [Wang, 1982](#)) has been useful in comparing the shape contours determined using 2DVD measurements made

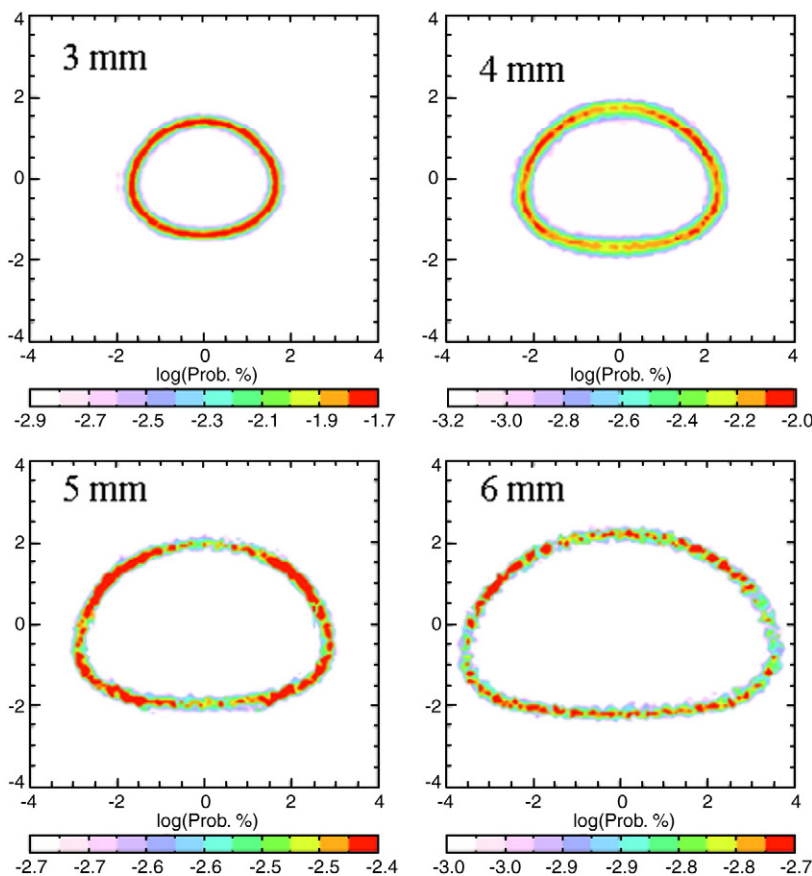
in natural rain reported in subsequent publications (e.g. [Huang et al., 2007](#)).

### 7.2. Shape comparison with wind-tunnel measurements

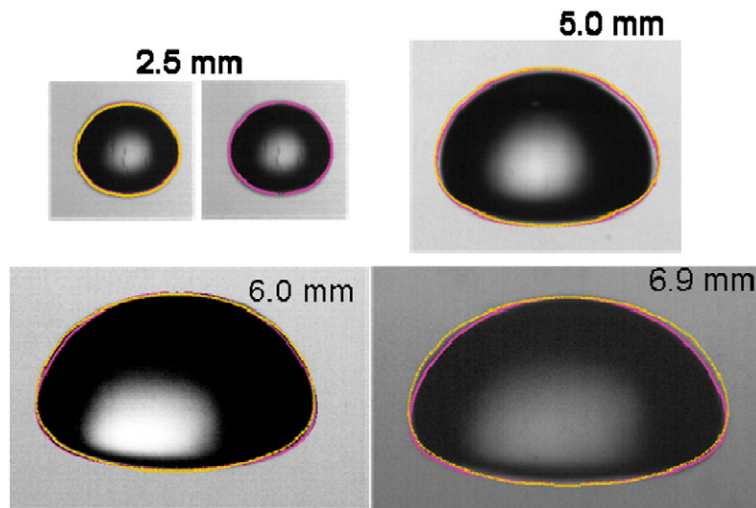
[Thurai et al. \(2009b\)](#) have compared the mean drop shape measurements made between the 2DVD and the wind-tunnel experiments. [Fig. 22](#) compares the two for  $D_{eq} = 2.5, 5.0, 6.0$  and 6.9 mm. The photographic images are from the wind-tunnel data whereas the pink curves represent the mean shape contours from the conical equation based on 2DVD data from the 80 m fall experiment. The agreement is remarkably close. Also shown in yellow are the contours from the [Beard and Chuang \(1987\)](#) numerical (B-C) model, derived using their Table 4. Here also the agreement is close. Note the 6 mm drop shape from the 2DVD and the wind-tunnel measurements show a flattening of the base, which is also predicted by the B-C model.

### 7.3. Canting angles from the 80 m fall (artificial rain) experiment

The term ‘canting angle’ ( $\beta$ ) was coined by the radio-wave propagation community. It is defined as the angle between (i) projection of the drop symmetry axis on the plane of polarization; the latter is the plane orthogonal to the direction



**Fig. 21.** Shapes of 3–3.25 mm, 4–4.25 mm, 5–5.25 mm and 6–6.25 mm drops from the 80 m fall experiment. The frequency of occurrence is contoured on log (probability) color scale determined from the 2DVD camera images. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



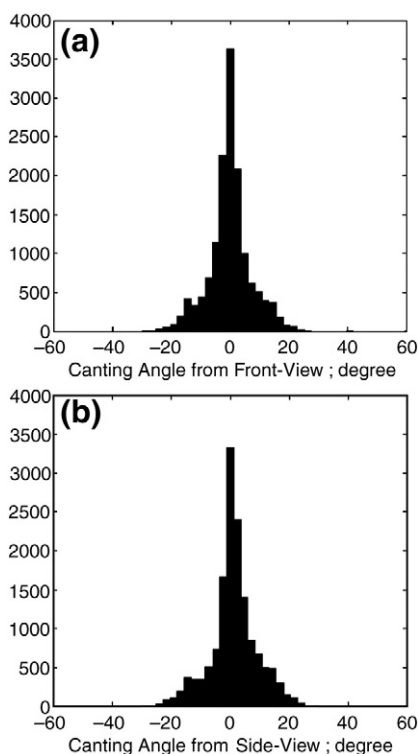
**Fig. 22.** Drop shapes from wind-tunnel measurements (photographic images) and comparisons with the conical fit to 2DVD data shown in pink (Eq. (1), [Thurai et al., 2009b](#)), and the [Beard and Chuang \(1987\)](#) model contour shown in yellow. The drop diameters correspond to  $D_{eq}=2.5, 5.0, 6.0$  and  $6.9$  mm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of incidence of the radio wave, and (ii) the projection of the local vertical direction on the polarization plane.

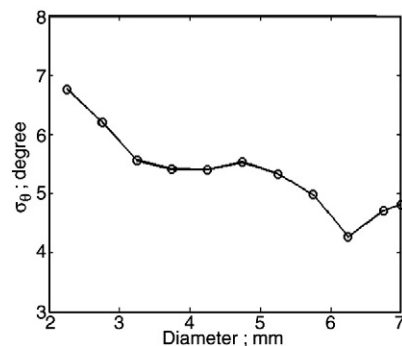
Theoretical studies by [Beard and Jameson \(1983\)](#), based on raindrop response to shear produced by homogeneous isotropic turbulence, predicted that the mean canting angle should be close to  $0^\circ$  with standard deviation ( $\sigma_\beta$ )  $< 4^\circ$  for turbulent

intensities  $< 0.2 \text{ m}^2 \text{ s}^{-3}$ . They found that  $\tan(\beta)$  was Gaussian but for small  $\beta$ ,  $\tan(\beta) \approx \beta$  and the Gaussian shape also applies to  $\beta$ . Their model predictions are valid above the surface layer and are consistent with the radar observations of [Hendry et al. \(1987\)](#). The latter work was based on definitive X-band radar measurements using rotating linear polarizations in heavy rain by [Hendry et al. \(1987\)](#) who showed that the mean canting angle ( $\beta$ ) was nearly  $0^\circ$  with standard deviation ( $\sigma_\beta$ ) close to  $6^\circ$  (see, also, Chapter 3 of [Brangi and Chandrasekar, 2001](#)).

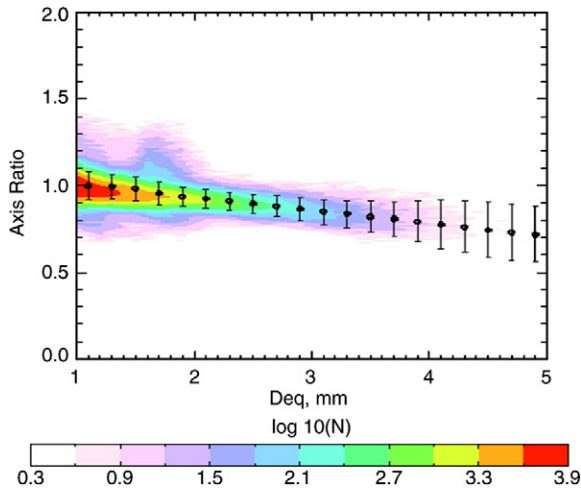
In the context of the 2DVD measurements, the term canting angle is used (even though it is defined for radio wave applications) since each camera ‘view’ can be thought of as being in the ‘polarization’ plane of a radar beam at zero elevation angle. As mentioned in [Section 5](#), the 2DVD gives two orthogonal views of the drop from which two estimates of the canting angle can be derived (one from each view). The term canting angle, with respect to the 2DVD, is the angle ( $\beta$ ) between the drop’s symmetry axis and the vertical direction



**Fig. 23.** Histograms of canting angles from (a) front (camera A) and, (b) side (camera B) views. Note that they are almost symmetric with mean of  $0^\circ$  and standard deviation  $7.2$  and  $7.8^\circ$ , respectively.



**Fig. 24.** Standard deviation ( $\sigma_\theta$ ) of  $\theta$  versus drop size ( $D_{eq}$ ) from the 80-m fall bridge experiment. The size intervals are from 2 to 7 mm with 0.5 mm step. The last data point represents those drops greater than 7 mm. Note that in calm conditions prevalent during the experiment, the large drops are more stably oriented (smaller  $\sigma_\theta$ ) than small drops (larger  $\sigma_\theta$ ). The 95% confidence interval of  $\sigma_\theta$  is estimated to be  $< 0.3^\circ$  for the range of diameters considered.



**Fig. 25.** Axis ratio versus drop diameter ( $D_{eq}$ ) where the frequency of occurrence is contoured on a log(number) color scale determined from the 2DVD measurements, together with the mean and  $\pm 2\sigma$  from the axis ratio measurements from the 80 m fall experiment in black. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

as obtained from each smoothed contour after deskewing; the distortion of the contour is caused by finite horizontal velocity of the drop. After deskewing, the two ‘intrinsic’ canting angles can be estimated (Schönhuber et al., 2008; Huang et al., 2008). From these two angles, the ‘true’ orientation of the symmetry axis of the drop can be calculated, as given by the polar angle ( $\theta$  from the vertical Z-axis) and the azimuthal angle ( $\varphi$ ).

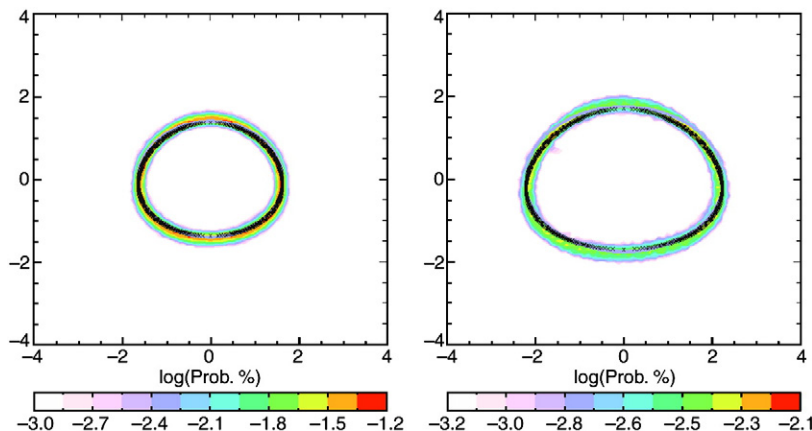
Huang et al. (2008) recently determined the histograms of canting angles for  $D_{eq}$  in the 2–8 mm range from the 80 m fall experiment referred to in Section 6. The canting angle histograms so obtained (see Fig. 23) showed that they were symmetric about a mean of  $0^\circ$  and that they had a standard deviation of  $7\text{--}8^\circ$ . Note the measurements were made during calm, low-wind conditions. From the two canting angles, the histograms of the polar angle ( $\theta$ ) and azimuthal angle ( $\varphi$ ) were also derived, the latter distribution being nearly uniform in  $[0, 2\pi]$ . Fig. 24 shows the plot of  $\sigma_\theta$  versus  $D_{eq}$  to

illustrate the fact that larger drops are expected to be more stably oriented than the smaller ones. Note that  $\sigma_\theta$  is approximately equal to  $\sigma_\beta$  (for values  $< 15^\circ$ ; Metcalf, 1988).

## 8. 2D-video disdrometer measurements in natural rain

A large set of 2D-video disdrometer measurements of drops in natural rain has been collected from a variety of locations, e.g., Okinawa, Sumatra, Toronto, Huntsville and Brisbane (Huang et al., 2007; Thurai et al., 2009a). Fig. 25 shows the axis ratio distributions versus drop diameter from the 2DVD measurements in natural rain taken near Brisbane in South-East Queensland, Australia. The frequency of occurrence is contoured on a log(probability) color scale. The variation is compared with the corresponding variation of the mean and two standard deviations (i.e.,  $\pm 2\sigma$ ) from the 80 m fall experiment (shown in black with error bars). Even though the measurements are given for  $D_{eq}$  down to 1 mm, it should be noted that below 1.5 mm, the 2DVD measurements will have decreasing accuracy due to the finite pixel resolution ( $\sim 0.2$  mm). Above 1.5 mm, the axis ratio measurements generally agree well with the 80 m fall data. However, note that in the 1.6–1.8 mm range, the axis ratios measured in natural rain show an unusually wide distribution, with a significant positive skewness. Similar observations have also been made with 2DVD measurements in Huntsville, Alabama (not shown here). Based on several types of calibrations involving precisely-machined metal spheres (see for example Fig. 3 of Thurai and Bringi, 2005), it seems unlikely that the instrumentation ‘spreading effect’ would contribute to the wide distribution seen in the 1.6–1.8 mm region. Instead, it is more likely that the wide distribution is evidence of oscillations due to the resonance with eddy-shedding frequency as, for example, discussed in relation to Fig. 12 in Section 4. In other diameter ranges, the 2DVD axis ratio measurements show almost the expected distributions.

In terms of drop shapes, Figs. 26 and 27 show the 2DVD data taken in Okinawa (a sub-tropical oceanic site) and Koto Tabang, Sumatra (equatorial site), respectively. They can be compared with similar drop shapes shown in Fig. 21 (top two panels) from the 80 m fall bridge (artificial rain) experiment.



**Fig. 26.** As in Fig. 21 except from 2DVD data from Okinawa, Japan during a ‘Baiu’ front event (Bringi et al., 2006). The  $D_{eq}$  is in the range 3–3.25 mm (left panel) and 4–4.25 mm (right panel).

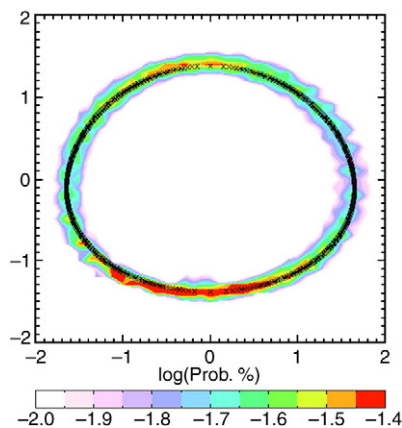


Fig. 27. As in Fig. 26 except from 2DVD data taken in Sumatra for the 3–3.25 mm drops.

One can note subtle differences in Fig. 26, especially the right panel versus Fig. 21 (i.e., the vertical chord is slightly larger than the horizontal chord leading to slightly larger axis ratios compared with the 80 m fall bridge data).

Further, Fig. 28 shows the histogram of the axis ratios for drops in 2–2.5 mm interval for the Sumatra data (solid green) compared with what we term as the ‘reference’ histogram from the 80 m fall (artificial rain) experiment for the same  $D$  interval (in magenta). The slight skewness in the Sumatra histogram (relative to the artificial rain) is a notable feature which tends to be accentuated for larger sized drops.

For example, Fig. 29 shows histograms of axis ratio within one convective event from Huntsville, AL (Thurai et al, 2009b) stratified by the initial strong convective phase when the rainfall exceeded 50 mm/h several times (marked as  $T < 06:30$  UTC in red) versus the later phase where the rainrate exceeded 50 mm h<sup>-1</sup> only once (marked as  $T > 06:30$  UTC in blue). The latter shows axis ratio distribution very similar to that found in the artificial rain experiment (see Thurai and Brangi, 2005), which gave a mean value of 0.815 for the 3.5–3.75 mm drop diameter interval. This value is close to the mode of the blue histogram corresponding to the second part of the event, but is significantly lower than the histogram mode during the first part of the event. The shift in the axis

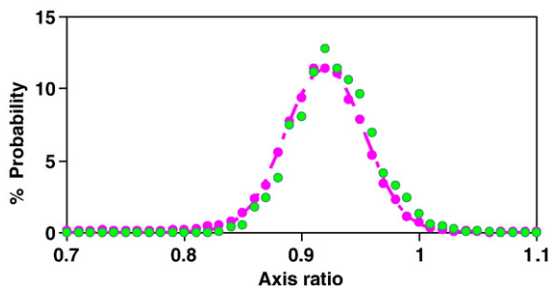


Fig. 28. Axis ratio distribution shown as green ‘solid circles’ for 2–2.5 mm diameter interval for the Sumatra data. The magenta points are the reference data from the artificial rain experiment (Thurai and Brangi, 2005). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

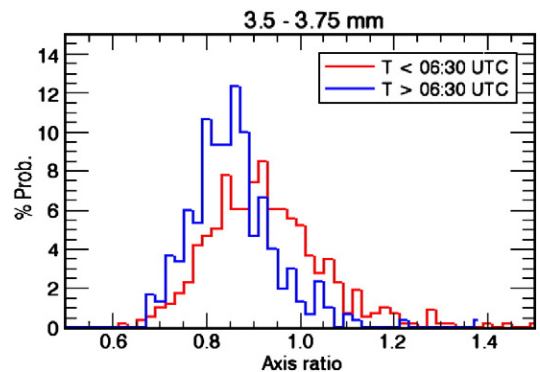


Fig. 29. Histograms of axis ratios for the 3.50–3.75 mm drop diameter range using the 2DVD data for the 25 August 2007 convective event in Huntsville Alabama. The beginning high rainrate (rainfall exceeding 50 mm/h several times) phase of this event is in red ( $T < 06:30$  UTC) and the later, lower rainrate phase (rainfall exceeding 50 mm/h once) is in blue ( $T > 06:30$  UTC). Note the histogram during the lower rainrate phase was found to be very similar to the axis ratio distributions from the 80 m fall (artificial) rain experiment, whereas the histogram during the high rain rate phase shows significantly higher axis ratios (relatively more spherical shapes) and a wider distribution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ratio distributions during the high rain-rate phase (both the higher value of the mode, the broader distribution as well as the skewness) towards the less oblate (or, “more spherical”) shapes may represent mixed-mode oscillations (similar to the drop shapes in the grey squares in Fig. 3 earlier). Note, however, this does not apply to the lower rain-rate phase of this event. The dual-polarized radar measurements from this event will be described later in Section 10.

## 9. Dual-polarized radar: microphysical information

Dual-polarized radars exploit two properties of the rain medium, (i) the mean oblateness monotonically increasing with increase in drop diameter, and (ii) the orientation of the symmetry axes being close to the vertical direction with a narrow distribution of canting angles. As a consequence of the latter, raindrops form a highly oriented medium whose eigen-polarizations are very close to the horizontal/vertical states. Thus, dual-polarized radars that transmit H and V polarizations are “tuned” to the rain medium. In a propagation sense, the waves attenuate and are phase shifted differently between H and V states (termed as differential attenuation and differential propagation phase). In the back scatter sense, the H-polarized reflectivity exceeds the V-polarized reflectivity, this increase being related to the oblateness (or, axis ratio) of the drops in the resolution volume. In addition, the canting angle distribution increases the cross-polarized reflectivity relative to the co-polarized reflectivity.

The principal radar measurements are (i) reflectivity at H-polarization ( $Z_h$ ), (ii) the differential reflectivity ( $Z_{dr}$ ) which is the ratio of reflectivities at H and V polarizations, (iii) the specific differential phase ( $K_{dp}$ ) which is proportional to the real part of the difference between the forward scattering amplitudes at H and V polarizations, (iv) the linear depolarization ratio ( $LDR$ ) which is the ratio of the cross-polar

reflectivity to the copolar reflectivity and (v) the correlation coefficient ( $\rho_{co}$ ) between the received copolar signals at H and V polarizations. For details of the various radar measurables we refer to Chapter 7 of [Bringi and Chandrasekar \(2001\)](#).

The characterization of the probability density function of the axis ratio ( $\alpha$ ) for a given  $D$ , or  $p(\alpha/D)$  is important for deriving estimates of the mass-weighted mean diameter  $D_m$  of the DSD (or, the median volume  $D_0$ ) and the rain rate  $R$  from radar measurements of principally  $Z_{dr}$  and  $K_{dp}$  (in addition, of course, to  $Z_h$ ). Henceforth we use  $r$  to refer to axis ratio (as opposed to  $\alpha$  used earlier).

The differential reflectivity ratio ( $Z_{dr} = Z_h/Z_v$ ) assuming perfect orientation is related to the reflectivity-weighted axis ratio  $r_z$  as ([Jameson, 1983](#)):

$$(Z_{dr})^{-1} \approx [r_z]^{7/3} \quad (3)$$

where  $r_z$  is defined via the drop size distribution,  $N(D)$ , and  $p(r/D)$  as:

$$r_z = \frac{\int D^6 N(D) \int r^{7/3} p(r/D) dr dD}{\int D^6 N(D) dD} \quad (4)$$

The above, along with Eq. (3) clearly shows that because of the  $D^6$  weighting, the pdf of axis ratio of the large drops dominate the differential reflectivity ratio (note also that axis ratio as defined here has  $r < 1$ ). Hence, it is important to characterize the 'tail' of the  $N(D)$  and hence the  $p(r/D)$  for large drops which are frequent in heavy rain. In general however, it is sufficient to characterize the mean and variance of  $p(r/D)$ . Note that the integral of  $rp(r/D)$  over its distribution is simply the expected value, i.e.,  $E(r/D)$ . The latter can be approximated as in Eq. (1) of the [Introduction](#) in linear form as  $E(r/D) \approx a - \gamma D$ , especially for  $D > 1$  mm. Hence, it follows that  $r_z = a - \gamma D_z$  where  $D_z$  is the reflectivity-weighted mean diameter or ratio of  $E(D^7)/E(D^6)$  where now expectation is over the  $N(D)$ . From Eq. (3), it follows that  $(Z_{dr})^{-1} \approx [a - \gamma D_z]^{7/3}$  and hence the radar measurements of  $Z_{dr}$  can be directly related to the  $D_z$  of the drop size distribution if the slope  $\gamma$  is fixed. For the gamma form of the  $N(D)$ , the  $D_z$  can be related to the median volume diameter ( $D_0$ ) via a function of the shape parameter ( $\mu$ ) only ([Ulbrich and Atlas, 1998](#)).

The specific differential phase ( $K_{dp}$ ) under perfect orientation can be expressed as:

$$K_{dp} = \lambda \int N(D) \int \text{Re}[\hat{h} \cdot \vec{f}(r, D) - \hat{v} \cdot \vec{f}(r, D)] p(r/D) dr dD \quad (5)$$

where  $\lambda$  is the radar wavelength and  $\vec{f}$  is the vector forward scattering amplitude. Under Rayleigh scattering, it can be shown that  $K_{dp}$  is only a function of the mass-weighted mean axis ratio ( $r_m$ ) and hence a full characterization of  $p(r/D)$  is not necessary ([Jameson, 1985](#)). The  $r_m$  is defined as:

$$r_m = \frac{\int D^3 E(r/D) N(D) dD}{\int D^3 N(D) dD} \quad (6)$$

where  $E(\ )$  stands for the expected value. Note that as before, if  $E(r/D) \approx a - \gamma D$ , then  $r_m = a - \gamma D_m$  where  $D_m$  is the mass-

weighted mean diameter of the DSD, i.e., the ratio of the 4th to 3rd moments of  $N(D)$ . Hence,  $K_{dp}$  can be directly related to  $r_m$  in the form ([Jameson, 1985](#)) as:

$$K_{dp} = C[180/\lambda]W(1-r_m) \quad (7)$$

where  $C$  is a constant,  $\lambda$  is the wavelength and  $W$  is the rain water content.

If the canting angle distribution is Gaussian with mean =  $0^\circ$  and standard deviation  $\sigma_\beta$ , then Eq. (5) is modified as:

$$K_{dp}(\sigma_\beta) = K_{dp}(0) \exp(-2\sigma_\beta^2) \quad (8)$$

The canting angle distribution as measured by the 2DVD has been described in [Section 7.3](#). If  $\sigma_\beta = 7^\circ$ , then the adjustment factor due to canting is approximately 3%.

If the transmitted polarization state is H-polarization, then the back scatter return from non-spherical and/or canted particles is composed of both a cross-polar signal (that is V-polarized) and copolar signal (that is H-polarized). The ratio of the corresponding reflectivities is defined as the linear depolarization ratio ( $LDR$ ) =  $Z_{vh}/Z_{hh}$ . Under Rayleigh scattering the  $LDR$  may be expressed as (for details refer to [Bringi and Chandrasekar, 2001](#)):

$$LDR = \left(\frac{1}{4}\right) [1 - \exp(-8\sigma_\beta^2)] [(1-r_z)^2 + \text{var}(r_z)] \quad (9)$$

The above equation clearly shows the dependence of  $LDR$  on the standard deviation of the canting angle distribution, the reflectivity-weighted mean axis ratio and the variance of the axis ratios. Of these three factors, the most important are  $r_z$  and  $\sigma_\beta$  with lesser effect of  $\text{var}(r_z)$ . In contrast to  $Z_{dr}$  and  $K_{dp}$ , the  $LDR$  will depend on both the mean and variance of the reflectivity-weighted axis ratio.

The final radar measureable we consider is the copolar correlation coefficient ( $\rho_{co}$ ) which is the correlation coefficient between the copolar signal returns. Factors that reduce the correlation coefficient from unity (for spheres or for scatterers fixed in space) are related (among other factors) to the shape variation across the DSD. For Mie scattering it is also reduced by changes in the back scatter differential phase across the DSD. However, for Rayleigh scattering and perfect orientation, the magnitude of  $\rho_{co}$  may be expressed as:

$$|\rho_{co}| \approx 1 - \left(\frac{1}{2}\right) \frac{[\text{var}(r_z)]}{[r_z^2]} \quad (10)$$

For example, if  $r_z = 0.8$  while  $\text{var}(r_z) = 0.005$ , then  $|\rho_{co}| = 0.996$ . Further reduction is possible if the canting angle distribution is considered. We note that an estimator of  $\sigma_\beta$  is possible by combining  $LDR$ ,  $Z_{dr}$  and magnitude of  $\rho_{co}$  ([Jameson and Davé, 1988](#); used later in [Section 11](#)).

From 2DVD measurements of  $N(D)$  we can compute the above radar measurables using T-matrix-based scattering calculations assuming, either (i) the mean axis ratio versus  $D$  fit to the 80-m fall bridge data shown in [Fig. 20](#) (black solid circles) along with Gaussian form for the canting angle distribution with mean = 0 and  $\sigma_\beta = 7.5^\circ$  (see [Section 7.3](#)), or

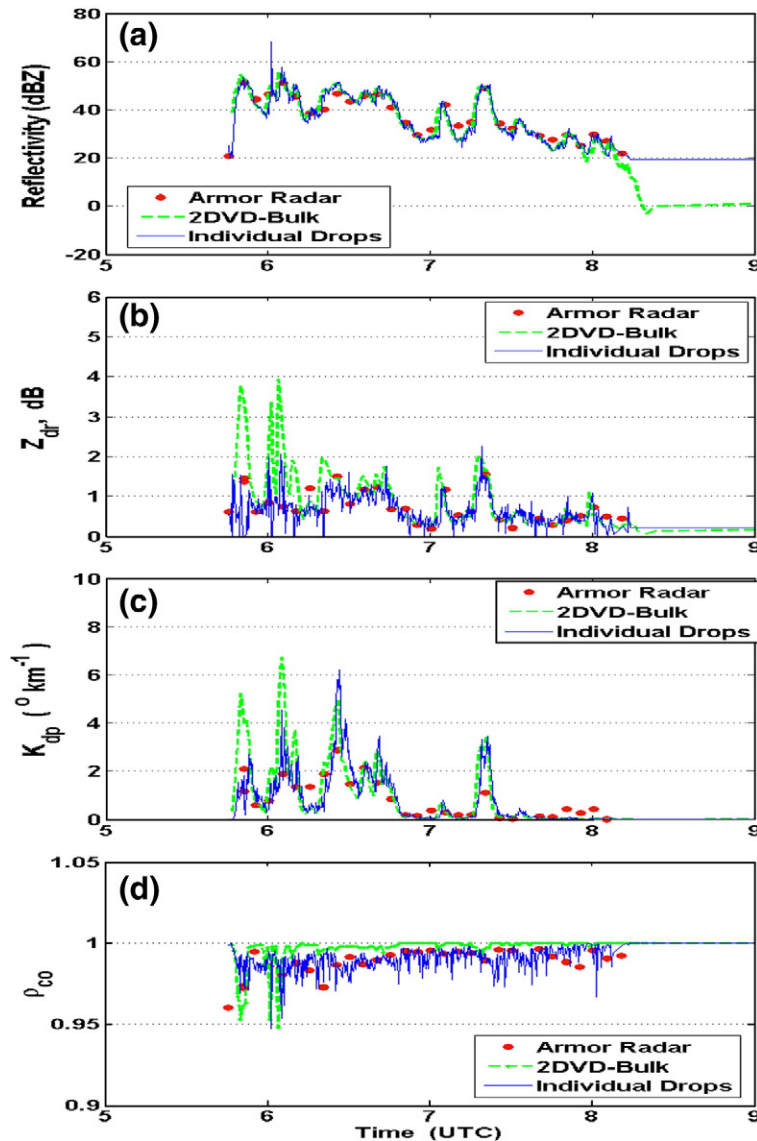
(ii) drop-by-drop information on shape (three cone shape parameters for each image) and the orientation angles ( $\theta, \varphi$ ) mainly for  $D > 1.5$  mm; for smaller drops (i.e.,  $D < 1.5$  mm), the  $E(r/D)$  from the laboratory data shown in Fig. 12 is used for the mean axis ratios with zero canting. Method (i) has been used by many researchers under different axis ratio and  $N(D)$  models, and with or without any canting angle distribution (see references in Chapter 7 of Brangi and Chandrasekar after whereas the drop-by-drop method is recent (Thurai et al., 2009b) and only possible with the 2DVD).

# 10. Axis ratio measurements in natural rain together with polarimetric radar observations

Very recently, drop axis ratio measurements in natural rain have been made in several locations using the 2DVD in

conjunction with simultaneous observations from polarimetric radar. Amongst these are (i) Huntsville, Alabama and (ii) South-East Queensland, Australia. Ongoing work in (ii) involves a synergistic approach to combine the 2DVD measurements and simultaneous observations using the CP2 facility in south-east Queensland comprising a dual-polarization S-band radar together with a precisely-aligned X-band radar (with cross-polar measurement capability).

The other location, viz. Huntsville, Alabama (USA), has a C-band polarimetric radar (named ARMOR, see Petersen et al., 2007) which has been used to make observations over the 2DVD site, 15 km away. An event analysis for an intense storm utilized the 2DVD images of individual drops which were processed to determine their individual shape and orientation, which in turn were used as input to the T-matrix method to compute the forward and backscatter amplitudes



**Fig. 30.** Time series comparisons of (a)  $Z_h$ , (b)  $Z_{dr}$ , (c)  $K_{dp}$  and (d)  $\rho_{co}$  for the 25 August 2007 event. In all cases, the blue line represents the calculations utilizing the individual drop information and the green line uses the 1-minute integrated DSDs with bulk assumptions regarding drop shapes and orientations. The red dots show the C-band radar measurements (located 14.5 km away) extracted from operational PPI scans and, with weighted 9-point average over the 2DVD site.

at C-band. The polarimetric radar variables were then calculated using the individual drop contributions over a finite time period, e.g. 1 min. In conjunction with the simultaneous C-band polarimetric observations, the synergistic approach showed that during the beginning period of the storm, the differential reflectivity ( $Z_{dr}$ ) and the specific differential propagation phase ( $K_{dp}$ ) showed lower than expected values, the lower values being consistent with the 2DVD based ‘drop-by-drop’ calculations. The comparisons are shown in Fig. 30 where the ARMOR radar measurements (over the 2DVD site) are shown as red dots and the 2DVD based calculations using the individual drop information are shown as a blue line. Over-plotted as green dotted lines are the calculations using 1 minute DSDs and bulk assumptions regarding the shape and orientation of drops, namely that each drop has the ‘most probable’ shape depending on its  $D_{eq}$  and that the drops have a symmetric canting angle distribution with zero mean and standard deviation of  $7.5^\circ$ . Whilst the reflectivity curves do not show much difference between the two cases, there are differences for the polarimetric radar parameters, as follows:

- (a)  $Z_{dr}$  and  $K_{dp}$  show significant differences at the beginning of the event
- (b) Copolar correlation coefficient ( $\rho_{co}$ ) shows differences throughout the event, with the ‘drop-by-drop’ method showing better agreement with ARMOR data.

Regarding point (a) above, the differences in  $Z_{dr}$  and  $K_{dp}$  can be attributed to drop shapes being ‘modified’ from the mean shapes at the beginning of the event. Referring back to Fig. 29 which compared the axis ratio distributions for the 3.5–3.75 mm drops for the first hour of the event (shown in red) and the following 2 h of the event (blue), we see clearly the distribution being shifted at the beginning of the storm and well as being somewhat more broadened. This example illustrates the importance of the drop axis ratio measurements in various rainfall regimes and climatologies.

## 11. Polarimetric radar-based estimates of canting angles

Very recently, it has been possible to utilize an advanced polarimetric radar facility to make inferences on drop canting angles. The radar facility referred to here is the S-band CSU-CHILL radar (Bringi et al., 2009) based in Colorado, USA, which uses a newly-installed 9 m dual-offset Gregorian antenna with an exceptionally high cross-polar performance. The antenna enables measurements of linear depolarization ratio ( $LDR$ ) down to almost  $-43$  dB. The radar was operated in the alternating H–V mode with 2 transmitters ‘firing’ alternately with the co-polar signals routed to the co-polar receiver and similarly with the cross-polar signals routed to a second receiver. The method of estimating  $\sigma_\beta$  from polarimetric radar measurements is from Jameson and Davé (1988); see, also, Bringi et al. (2008) and the references therein. Briefly, the assumptions are Rayleigh scattering and Gaussian form for the canting angle distribution with mean  $0^\circ$  and standard deviation  $\sigma_\beta$ . The  $\sigma_\beta$  is derived from the measured values of  $Z_{dr}$ ,  $LDR$  and  $|\rho_{co}|$ . In essence, the terms involving  $r_z$  and  $\text{var}(r_z)$  in Eq. (9) are estimated from  $Z_{dr}$  and  $|\rho_{co}|$  (see Eqs. (3) and (10)). The data used were from one low elevation angle sweep

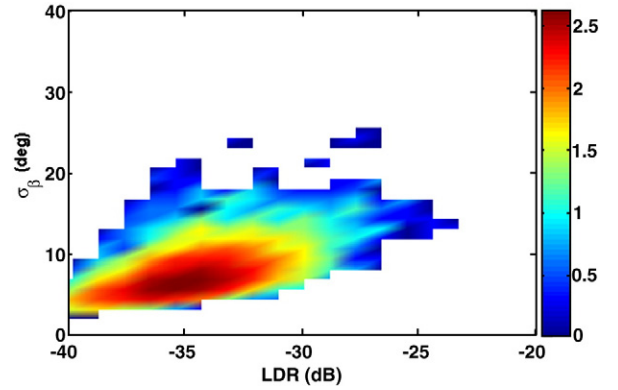


Fig. 31. Contoured frequency of occurrence of the estimated  $\sigma_\beta$  with  $LDR$  from the ‘CSU-CHILL’ radar measurements in light-to-moderate stratiform rain. The color scale indicates the relative frequency of occurrence in log scale.

at short ranges ( $5 < \text{range} < 30$  km with 150 m gate spacing) in stratiform rain with embedded convection which occurred on 5 June 2008. The histogram of  $LDR$  for this dataset showed that the mode for stratiform rain was  $-35$  dB; the system  $LDR$  limit was estimated to be close to  $-40$  dB with some data points lying as low as  $-43$  dB. Fig. 31 shows a color-scale plot of  $\sigma_\beta$  (the physical interpretation was given earlier in Eq. 9). The figure (representing contoured frequency of occurrence) shows that as  $LDR$  increases the drops have a larger  $\sigma_\beta$ . The most probable values in this case of stratiform rain with embedded convection was found to be close to  $[-35 \text{ dB}; \sigma_\beta = 7^\circ]$ . Fig. 32 is the histogram of  $\sigma_\beta$ ; it shows the mode at around  $7^\circ$ , with a significant positive skewness. Around 16% of occurrences exceed  $10^\circ$  and 3.3% of the occurrences exceed  $15^\circ$ . These radar-based estimates of  $\sigma_\beta$  are consistent with the 2DVD measurements as deduced from Fig. 24 in Section 7.3. Further, the radar-based frequency of occurrence of  $\sigma_\beta > 10^\circ$  is supported by 2DVD measurements in natural rain under moderate wind conditions, which had yielded values of around  $12^\circ$  (Thurai et al., 2009b).

Another point which arose from the above study concerns the stability of larger drops compared with smaller ones. Fig. 33 shows the variation of  $\sigma_\beta$  determined from the S-band polarimetric radar data versus the differential reflectivity,  $Z_{dr}$ , from the same stratiform event mentioned earlier. The color

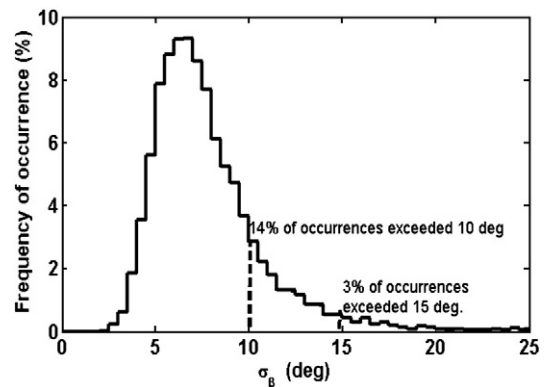


Fig. 32. Histogram of the estimated  $\sigma_\beta$  from radar measurements taken from the same radar measurements as in Fig. 31.

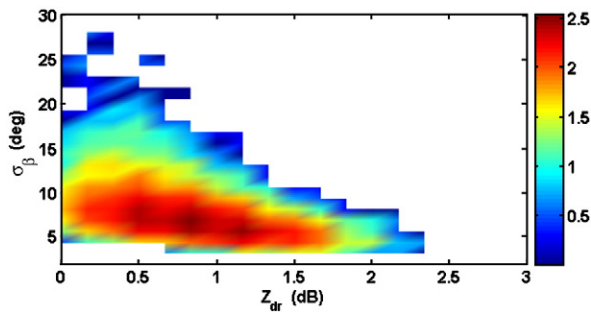


Fig. 33. Variation of the standard deviation ( $\sigma_\beta$ ) of the canting angle with S-band differential reflectivity ( $Z_{dr}$ ) determined from the CSU-CHILL radar observations during a stratiform rain event.

scale represents the number of cases on a log scale. A general decrease of  $\sigma_\beta$  with  $Z_{dr}$  can be seen, indicating that larger drops – the presence of which contribute to larger  $Z_{dr}$  – have smaller spread in their canting angles. This is consistent with the 2DVD measurements of canting angle histograms for various drop diameters which showed the standard deviation of the zenith angle decreasing with  $D_{eq}$ , as shown earlier in Fig. 23. Both Figs. 23 and 24 indicate that the larger drops are more stably oriented than the smaller ones.

## 12. Conclusions from 2DVD measurements

To summarize the recent work on 2DVD measurements, the capability of the instrument for accurate contouring of drop shapes and for determining drop canting angles has been established from the 80-m fall (artificial rain) experiment conducted under calm conditions. Since then, measurements in natural rain have begun in several locations, including Alabama (USA), Brisbane (Australia), Sumatra (Indonesia) and Okinawa (Japan). In the former two cases, the 2DVD measurements are being made in conjunction with polarimetric radars. Data analysis is ongoing in order to identify meteorological conditions in which significant differences in drop shapes may occur. So far, one event has been identified which showed a significant shift in the 3 mm axis ratio distributions during the initial period of the event. Further work will examine axis ratio distributions in deep convective storms and contrast these with stratiform events. Work on drop canting angles will also continue using 2DVD measurements in natural rain as well as inferences made using the CSU-CHILL radar facility in moderate and intense convection. The work will be an extension of the already estimated canting angle distributions for low-to-moderate rainfall events, under low wind conditions.

To supplement the drop shape studies, collision-induced drop oscillations will be examined using the Mainz wind-tunnel facility. This extension of the laboratory work begun by Hans Pruppacher at UCLA in the late 1960s will evaluate how collisions between raindrops and much smaller drops affect the average and variance of raindrop axis ratios.

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